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Classifying YouTube channels: Algorithmic approach to channel mapping

by Anna Zaitsev and Sam Clark

Abstract

YouTube is one of the most influential platforms for news and political commentary; however, its vast scale makes it challenging to study the full range of content and communities that it hosts. Most existing research focuses only on the most popular channels, overlooking the large number of smaller creators who may reach sizable and politically engaged audiences. In this study, we introduce a new approach to identifying and classifying YouTube channels based on patterns in who subscribes to them. Using this method, we mapped the landscape of English-language political channels on YouTube during 2020. Our findings indicate that prior studies have significantly underestimated the prevalence of right-leaning and conspiracy-themed content, which is more prevalent among smaller channels. We also show that large media outlets tend to dominate left-leaning content, while right-leaning content is primarily produced by independent creators. This work demonstrates the value of scalable, data-driven methods for understanding online media ecosystems and offers a more complete picture of the political content available to YouTube users.

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I. Introduction

With over two billion monthly active users, YouTube is one of the world's largest social media platforms and the largest video-sharing platform. Initially founded in 2005 and acquired by Google in 2006 (Burgess and Green, 2018), the platform was popularized as a video-sharing resource for independent content creators. Content creators utilize the platform to produce a diverse range of content, including political commentary, makeup tutorials, music videos, vlogs, how-to videos, and gaming content. Nearly every topic and category imaginable is covered on the platform, including content from traditional television and network media outlets. Users have embraced YouTube as their source of entertainment and news.

According to a recent study by the Pew Research Center, 26 percent of adults in the United States obtain news from YouTube (Stocking, *et al.*, 2020). As YouTube has grown, its scale has presented challenges in studying the ideas being shared and communities being formed on the platform.

On cable and network television, individuals can be aware of all the channels available in a region and have a sense of the type of content common to them. Additionally, Nielsen, a company specializing in information and data, tracks television ratings, which can be used to compare audience sizes for different types of content. With millions of channels on YouTube, an individual (or even a group of researchers) can't collect similar data manually.

Prior studies have taken two different approaches, focusing on the breadth or depth of YouTube data. Studies focusing on the breadth of YouTube attempt to analyze the content (for example, political channels) on a set of the most popular channels (Stocking, *et al.*, 2020; Dinkov, *et al.*, 2019). This limitation means that a significant number of smaller channels, which in aggregate could have more traffic than popular channels, are excluded from the analysis. Conversely, other papers focus on the depth of individual communities and attempt to understand a discrete subset of YouTube content. These studies provide detailed insight into various niche communities (Ribeiro, *et al.*, 2020; Papadamou, *et al.*, 2021) but offer no comparative data regarding the proportionality of these niche communities. Additionally, researchers using both approaches have failed to measure the extent to which their studies cover the overall community or category content.

This study proposes a novel method, titled *chan2vec*, based on the *word2vec* machine learning algorithm (Mikolov, *et al.*, 2013), for automated channel discovery and classification. This enables the analysis of orders of magnitude more channels when studying communities or categories on YouTube. When applied, it can provide a more realistic figure for assessing the sheer number of channels belonging to different categories than prior methods applied.

With this method, one can investigate the reach and number of channels focusing on any topic as long as training data is available. However, due to YouTube's increasing importance as a news source (Stocking, *et al.*, 2020) and the number of datasets focusing on political YouTube content, we focus our experiments on YouTube channels that discuss political and cultural issues. Our example, which applies the classification method, provides a snapshot of YouTube's political channel landscape in the turbulent year of 2020. This year, social media platforms, including YouTube, experienced increased traffic due to restrictions and closures resulting from the COVID-19 pandemic, making it an interesting historical flashpoint from a social media research perspective (Tsao, *et al.*, 2021; Cho, *et al.*, 2023). While broader platform interventions and policy changes, such as content removal, demonetization, and algorithmic ad placements, have undoubtedly influenced channel visibility and survival over time, our analysis does not aim to trace these dynamics or explain longitudinal changes. Instead, it focuses on mapping the structure of a large and often under-sampled content ecosystem at a specific point in time.

Finally, this study expands on a dataset manually created for a study presented in Ledwich and Zaitsev (2020). The dataset created for that study, the Recfluence dataset (Recommendation Influence), classified 800 most watched sociopolitical channels manually. These classifications were utilized as the training data for the chan2vec method.

In sum, when compared to the prior dataset, the Recfluence mapping of most popular channels, the approach reveals that many more channels could be classified as right-leaning than left-leaning, even when left-leaning channels dominate the mainstream. Furthermore, it shows that certain problematic channel types, such as "White Identitarian" channels, are rare on YouTube.

II. Related work

YouTube communities have been analyzed from different perspectives and methodologies. Burgess and Green (2018), for instance, discussed the history of YouTube and the platform's important role in participatory culture, where users themselves produce content. They examined how traditional media, professional content creators, and niche communities utilized the platform and discussed the tensions between commercial and user-created content. A 2020 metadata-driven analysis revealed that YouTube was primarily dominated by channels with over 100,000 subscribers (Rieder, *et al.*, 2020). Entertainment and gaming channels were most numerous and most viewed on the platform, whereas channels concerning "society" gained only four percent of the views. However, "society" category videos, of which politics is a subsection, made up 24 percent of the channels with more than 1000 000 subscribers, indicating that content creators who engage in this type of content were very active (Rieder, *et al.*, 2020) even if the view count was marginal when the whole platform was investigated. Many studies have further explored the role of the political YouTube from various perspectives. It has been studied as a source of information among the youth (*e.g.*, Zimmermann, *et al.*, 2022), a medium of left-wing political thought (Cotter, 2022), or as a source of conspiracy theories (Faddoul, *et al.*, 2020) and right-wing radicalization (Ribeiro, *et al.*, 2020; Ledwich and Zaitsev, 2020; Hosseinmardi, *et al.*, 2021).

From a technical perspective, previous research on YouTube content has largely focused on classifying either videos or channels. Large video datasets made available by YouTube (Karpathy, *et al.*, 2014; Abu-El-Haija, *et al.*, 2016) have led to significant advances in machine-learning video classification (Karpathy, *et al.*, 2014; Chen, *et al.*, 2017; Huang, *et al.*, 2018). Several studies have utilized metadata and video comments for video classification tasks, such as identifying conspiracy videos (Faddoul, *et al.*, 2020), investigating 'incel' content (Papadamou, *et al.*, 2021), or mapping the entirety of the platform (Reider, *et al.*, 2020).

Video-level classification can also be aggregated into channel-level classification; many channels focus on a narrow topic. Manual classification of channels has been used to study right-wing content (Ribeiro, *et al.*, 2020) and the full spectrum of political content (Ledwich and Zaitsev, 2020; Stocking, *et al.*, 2020). Machine learning approaches have also been used to predict the political lean of a channel (Dinkov, *et al.*, 2019), to predict whether a channel covers socio-political content along with ideology (Laukemper, 2020), and by YouTube internally to predict broad categories (Simonet, 2013), thought the last relies on propriety data.

Beyond YouTube, various studies have also used natural language processing to classify political content on other platforms. For example, studies have experimented with predicting the political ideology of Twitter users (Preoţiuc-Pietro, *et al.*, 2017) and shown how one can identify biased, hyper-partisan news outlets based on writing style (Potthast, *et al.*, 2018). Furthermore, datasets have been compiled to provide information on news outlets and their reliability based on fact-checked news articles (Horne, *et al.*, 2018), which have then been utilized to predict the bias and reliability of news outlets (Baly, *et al.*, 2018).

Based on prior research, natural language processing seems like a promising tool for large-scale channel analysis. Next, we will outline our machine learning-based approach and explain how our predictive model finds new channels and assigns them thematic tags.

III. Data collection and chan2vec

1. Methods applied in prior studies

One challenge of analyzing communities or categories on YouTube is identifying channels that belong to them. YouTube does not provide granular categories or ways to browse categories extensively. Furthermore, even if the platform provided more granular filtering features, focusing exclusively on larger

channels would provide limited coverage of all content, given its scale.

In past studies of communities and categories on YouTube, two primary approaches have been used to identify channels to focus on:

1. Start with a set of seed channels, then manually review channels that YouTube recommends as similar or that have videos frequently recommended for videos on these seed channels. Add channels that are found to fit in the community or category of interest to this set and repeat the process until saturation is reached or resources are exhausted (*i.e.*, snowball sampling method) (Lee, *et al.*, 2006)

2. Create a set of keywords associated with the community or category of interest, search YouTube for these keywords, then manually review the videos and/or channels returned.

These approaches have two major problems. First, they require a manual review of channels, which is time-consuming and significantly limits the number of channels that can be found. Second, YouTube's search and recommendation systems are less likely to return channels with a small number of subscribers and channels that post content that comes close to violating the platform's terms of service (in some cases, they restrict these channels altogether).

Our data collection and classification process is summarized in below. First, we explain how data was collected, followed by a detailed description of the channel discovery channel classification process. This is followed by channel discovery performance analysis, which is included in the Appendix.

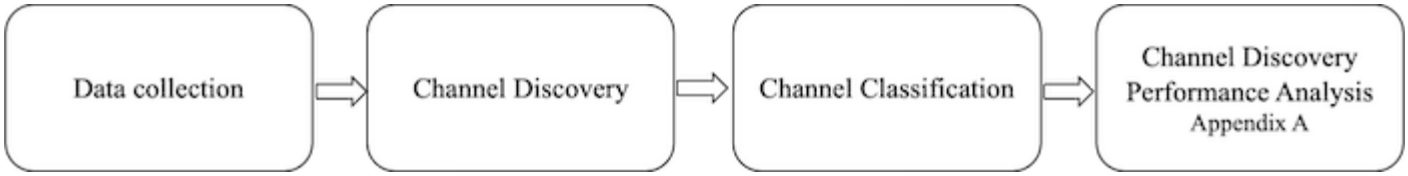

Figure 1: Channel classification process.

Furthermore, we apply two other datasets, the Laukemper data set (Laukemper, 2020) and the already-mentioned Recfluence dataset (Ledwich and Zaitsev, 2020), for data exclusion and classification. Utilization of both sets of data will be discussed in following sections.

2. Data collection

Our channel classification and discovery methods rely critically on the concept of video "commenter subscriptions." These are the channels to which a commenter subscribes, and for a large percentage of commenters, they were visible to the public on commenters' profile pages when this study was conducted in 2020. Figure 2 depicts a sample user subscription page.

To collect channel subscription data, we used the YouTube Data API in 2020, when public subscription lists were still widely accessible on user profile pages. Starting points for data collection were manually selected based on known socio-political channels, providing seed inputs for the iterative discovery process. All channels were consistently tracked using their unique 24-character channel ID strings, ensuring accurate matching and deduplication throughout. Unfortunately, since the data collection was conducted in 2020, YouTube has changed the visibility settings of the subscriptions, and the current platform is less conducive to this form of data collection. However, while prior studies have also used the YouTube API, our contribution lies in how this data is leveraged at scale through the chan2vec approach, enabling

significantly larger channel discovery and improved tagging performance compared to earlier methods.

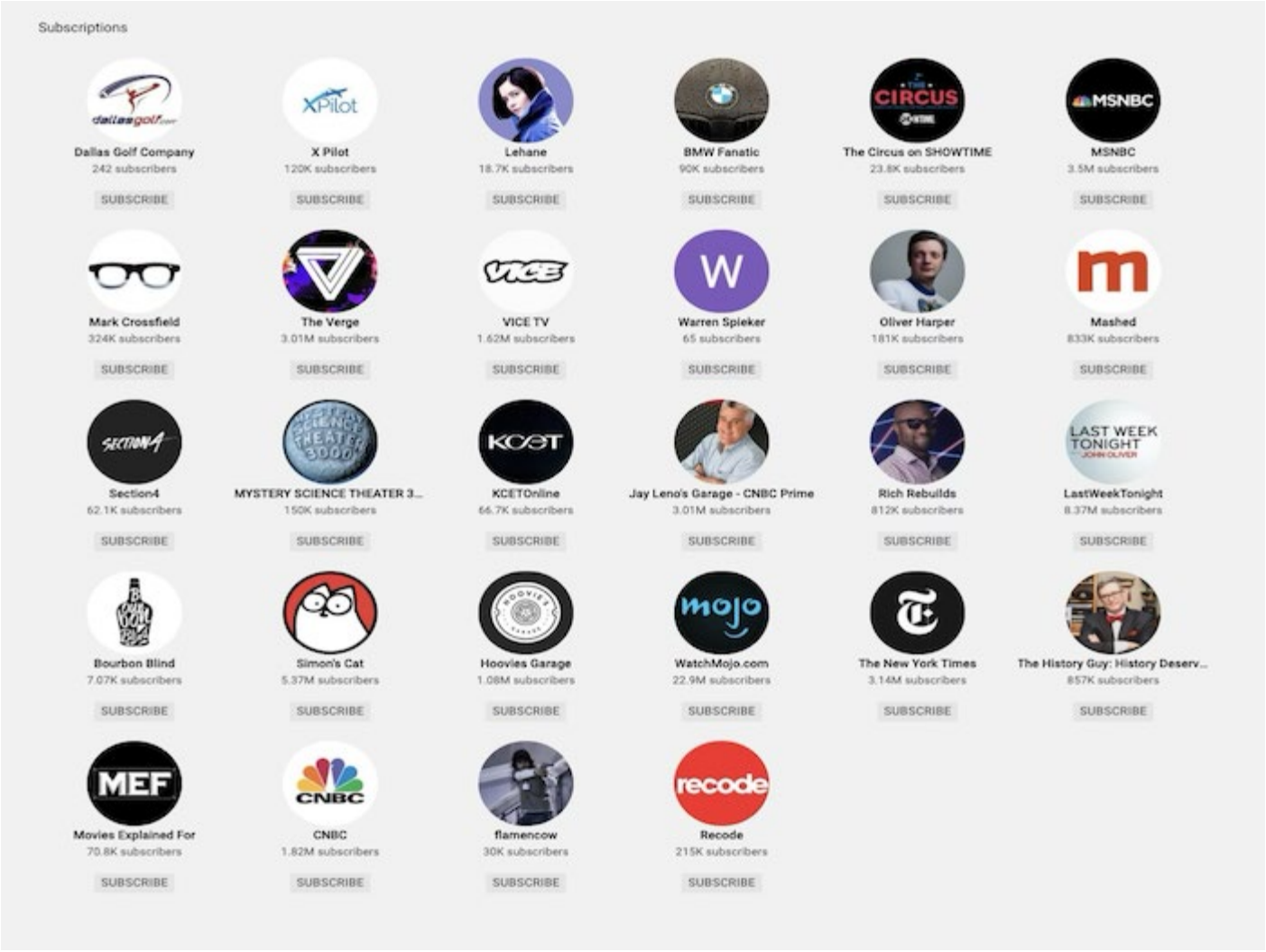

Figure 2: An example publicly available subscription page of a YouTube user.

To collect this data for a given channel, we proceed as follows:

- We query the 30 most recently uploaded videos for the channel.
- We query up to 100 comments for each of the 10 most viewed of these videos.
- We query subscription data for all of these commenters with public subscription pages (which we have found is roughly 30 percent of commenters).

First, we use a method that returns only 30 subscriptions if they are available. We call these "sample commenter subscriptions" since they frequently do not cover all channels to which a commenter subscribes. However, for up to 10 commenters with public subscription pages, we use a more computationally intensive method to query all available subscriptions. On average, a user has 210 subscriptions, which we designate as "full commenter subscriptions."

Commenter subscriptions can be used in various ways for channel classification or topic modeling. Our

approach was inspired by work done at Instagram to create account representations, embeddings, for a similar purpose (Medvedev, *et al.*, 2019). We decided to infer channel embeddings from commenter subscriptions using a word2vec-like embedding framework that we call chan2vec, short for 'channel to vector', since it applies vector-based embedding techniques to channels. Instead of representing each subscription list as a simple checklist of channel names, we use a technique inspired by natural language processing that learns patterns based on which channels tend to appear together across many users. This approach captures more subtle relationships between channels than traditional methods like Principal Component Analysis (PCA) or Latent Dirichlet Allocation (LDA) and helps group similar channels more effectively, even when their individual audiences differ. Typically, word2vec learns vectors (commonly referred as embeddings in this scenario) for words based on the context in which they appear across a corpus of sentences (Mikolov, *et al.*, 2013). An example of dyads is presented in Figure 3. Rome and Sparta are more similar to each other; the vectors created from these words are closer to each other than for other words. The learning algorithm aims to learn vectors such that semantically similar words have similar vectors (specifically, high cosine similarity).

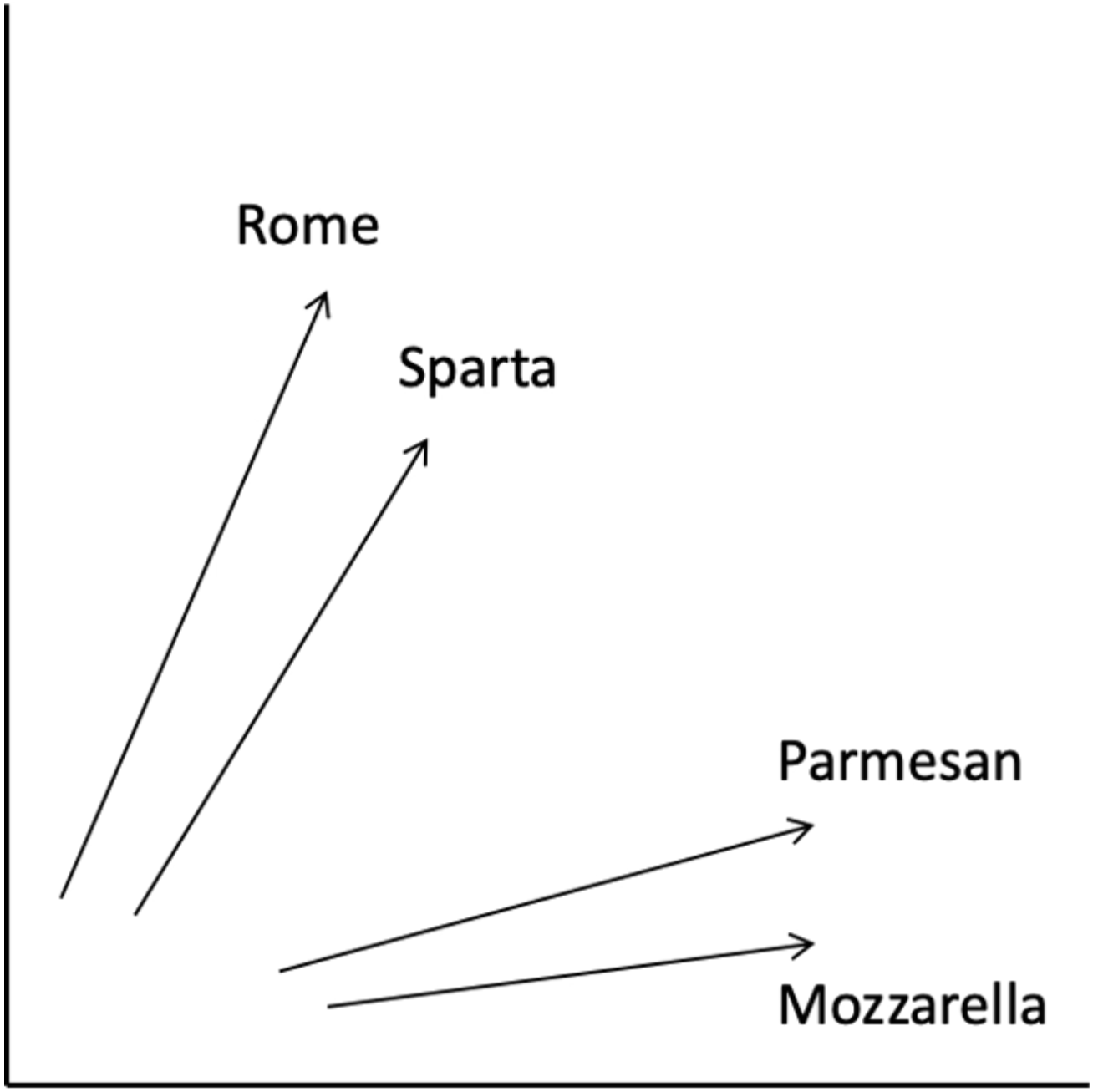

Figure 3: Example of word embeddings vectors.

For chan2vec, we treat each commenter's list of subscribed channels like a sentence, where each channel ID functions as a word. This allows us to apply techniques from natural language processing to learn how channels relate to one another. To learn useful embeddings for a channel, we must have subscription information for a sufficient number of subscribers to obtain the proper signal. We set the minimum commenter subscription threshold at five, but further analysis is necessary to determine the optimal threshold.

To generate the embeddings used for chan2vec, we create "sentences" for each unique commenter for whom we have collected subscription data. All channels from these "sentences" that occur fewer than five times in the dataset are filtered out. Next, all "sentences" that comprise fewer than three channels (meaning the commenter subscribes to fewer than three channels) are also filtered out. It is unclear how channels are ordered on public subscription pages, so we randomly shuffle the channels in "sentences" to eliminate any bias that the default ordering might introduce.

Unless otherwise specified, we apply continuous bag of words, a window size of eight, 200-dimensional vectors, and the other default word2vec parameters. Many of the experiments involve small datasets, so we cannot significantly optimize parameters without the risk of overfitting.

Our method for channel discovery is a simple extension of the chan2vec-knn and data collection methods described already. This method does not suffer from the limitations of the prior approach and enables the analysis of orders of magnitude in more channels than have been analyzed in previous studies. The method utilizes the fact that when querying commenter subscriptions for each new channel, data about many other channels with co-occurring subscriptions is also gathered. Due to the large number of subscriptions the average commenter has, new channels can be discovered rapidly. First, the method uses an iterative approach that casts a wide net to identify "candidate" channels or channels likely to belong to the category or community of interest. Then, it uses a more accurate model to predict which of the candidate channels belongs to the category or community.

Next, we use chan2vec to generate the embeddings, which are then used to make predictions on the potential channel tags with the help of K-nearest-neighbors (KNN) and a dataset of labeled channels. We use a KNN model to classify channels based on the labels of their closest peers in the embedding space. For example, to predict whether a channel is political, we look at the percentage of the K nearest channels labeled political. We refer the process of using KNN with chan2vec embeddings as "chan2vec-knn". As with word2vec, we use cosine distance to measure the distance between any two-channel embeddings.

After identifying candidate channels, we make final predictions for them. We use the 200-dimensional embeddings generated during the final round of channel discovery but also generate another set of 16-dimensional embeddings. We apply KNN separately using each of these and average the results to generate a slight improvement in accuracy and a smoother precision/recall curve. Finally, we filter out channels with fewer than 20 commenter subscriptions since all channels in the Recfluence dataset had more than 20 commenter subscriptions.

Our channel identification process discovered a vast number of channels:

- 12,600,000 total channels
- 1,600,000 with at least five or more commenter subscriptions necessary for chan2vec
- 600,000 of these with over 10,000 subscriptions (the threshold we set for the experiment)
- 25,000 of these were identified by the discovery process as candidate socio-political channels
- 7,000 of these are predicted to be socio-political

This large number of channels demonstrates why a machine-learning approach, such as chan2vec-knn, is necessary for classifying channels discovered through commenter subscriptions. The process is summarized in Figure 4.

A detailed description of the method, the two algorithms, and the process for creating the embeddings, along with a set of tests to validate the model, is presented in the Appendix. The Appendix provides a detailed discussion of the algorithms and the experiments that validate our model, including the set of channel discovery performance analyses. As stated earlier, the method itself is agnostic regarding the classification topics. One can use it to classify anything from music videos to makeup tutorial channels on YouTube. We use an example of sociopolitical channels because a large training data set was available, and the expansion of the data helps extend prior studies on the topic.

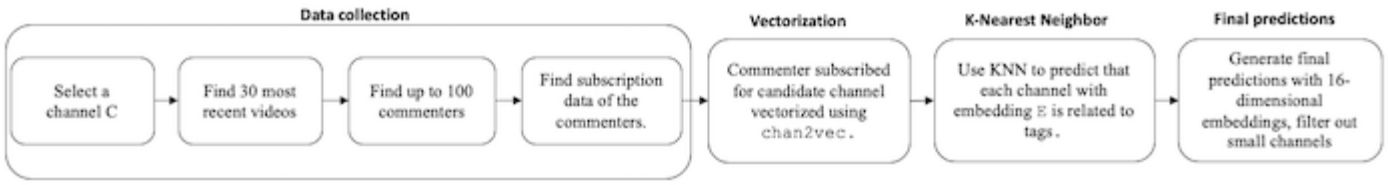

Figure 4: Data collection and prediction process.

Note: Larger version of Figure 4 available here.

IV. Classifying sociopolitical YouTube channels

Now that we have discovered channels that fit the criteria for the Recfluence dataset of socio-political channels, we classify further information about these channels. We use the media type and soft tag labels specified for channels in the Recfluence dataset. The Refluence dataset, which was created to study the recommendations algorithm and whether the recommendation algorithm is directing users to radicalizing content, is discussed in detail by Ledwich and Zaitsev (2020). The dataset labeling was created by three people who reviewed each channel manually to annotate them and apply a set of labels created during the labeling process as emergent themes. In this study, we originally used the Ledwich and Zaitsev (2020) study's set of 17 soft tags, *i.e.*, labels that describe the channel content type and three media types. Each channel can only have one media type but multiple soft tags. However, during the research, we discovered that many soft tags have a very limited number of examples in the Recfluence dataset, making it difficult to train and test models on them. Therefore, we limit ourselves to a subset of 12 soft tags with greater than 30 instances in the dataset. Table I provides a summary of the types and tags.

The media type tags differentiate between the content creators, distinguishing content created by independent YouTubers from content created by other outlets. The soft tags describe the political leaning of

the channel content and the high-level subject matter that the political channels engage in, such as focusing on singular issues, for example, men's rights activism, conspiracy theories, or current culture war topics. For longer descriptions, see Ledwich and Zaitsev (2020).

To predict soft tags and media types of channels, we use chan2vec-knn with the embeddings generated from the last round of the channel discovery experiment and the Recfluence tag labels as our labeled dataset. Previously, we used K=10, but due to the small number of channels with some labels, we used K=5 for this experiment. Metrics for this experiment can be found in Table VI, in the Appendix.

Table I: Media types and soft tags.

Media type	Description	Example channels
YouTube	Independent YouTube creators.	Timcast IRL, Jordan B Peterson, StephenCrowder, The Majority report w/Sam Seder
Mainstream Media	Content created by corporations such as cable news channels.	CNN Fox News, MSNBC, PBS News Hour, DW News, BBC News
Missing Link Media	Channels funded by companies or venture capital, but not large enough to be considered "mainstream". They are generally accepted as more credible than independent YouTube content.	Vox, The Young Turks, Vice News, cut, HuffPo
Partisan Left	Mainly focused on politics and exclusively critical of Republicans.	Late Show with Stephen Colbert, MSNBC, MeidasTouch, Vox
Partisan Right	Mainly focused on politics and exclusively critical of Democrats and supports Trump.	Fox News, Blaze TV, Young Americas' Foundation, Sky News Australia
	Regularly	

Conspiracy	promotes a variety of conspiracy theories.	The Next News Network, Anonymous Official
Anti-SJW	Have a significant focus on criticizing "Social Justice".	The Joe Rogan Experience, Timcast IRL, Ben Shapiro, The Podcast of the LotusEaters
Social Justice	Espouses progressive views on social issues with a focus on identity politics, intersectionality, and political correctness.	The Young Turks, The Majority report w/Sam Seder
White Identitarian	Believes in and identifies with the superiority of "whites".	AustralianRealist, Black Pidgeon Speaks
Anti-Theist	Self-identified atheist who are also actively critical of religion.	Secular Talk, Matt Dillahunty
Socialist	Focused on problems with capitalism.	Some Some More News, HasanAbi, Vaush
Religious Conservative	Focused on promoting Christianity or Judaism in the context of politics/culture.	Ben Shapiro, Matt Walsh, Glenn Beck
Libertarian	Focused on individual liberties and generally skeptical of authority and state power.	Hoover Institution, John Stossel
Educational	Has significant focus on educational material related to politics/culture.	Yad Vashem, Library of Congress
		BBC News, PBS

State funded	Channels funded by a government.	News Hour, Al Jazeera English

We first apply the chan2vec-knn tag classification approach from the previous section to the 7,225 newly discovered socio-political channels to obtain tag predictions for them. We then combine the new socio-political channels with the Recfluence channels and analyze the amount of traffic for all channels (new and from Recfluence) with a given soft tag. For the amount of traffic, we use Social Blade (2019) to obtain the number of views for each channel over the previous 12 months.

V. Findings and discussion

Our channel discovery method found nearly 10 times more socio-political channels than the Recfluence dataset. To better understand what channels were discovered, we break out the number of discovered channels by soft tag in Figure 4 and the Appendix. The number of channels increased for all soft tags but was particularly significant for “Partisan Right,” “AntiSJW,” and “Conspiracy.”

We know that a large number of the newly discovered channels are small. To better understand the impact of adding these channels, we must account for their popularity. We first do this by looking at the number of total channel subscribers with a given soft tag in Figure 5. We can see that the increase is still significant for most soft tags but not nearly as large as the number of channels would indicate. The most considerable increase occurs for “Conspiracy,” which goes from the fifth most popular soft tag to the second most popular soft tag. Notably, the “Partisan Left” category, which initially appeared dominant, drops to third place in popularity when we include smaller channels in the analysis.

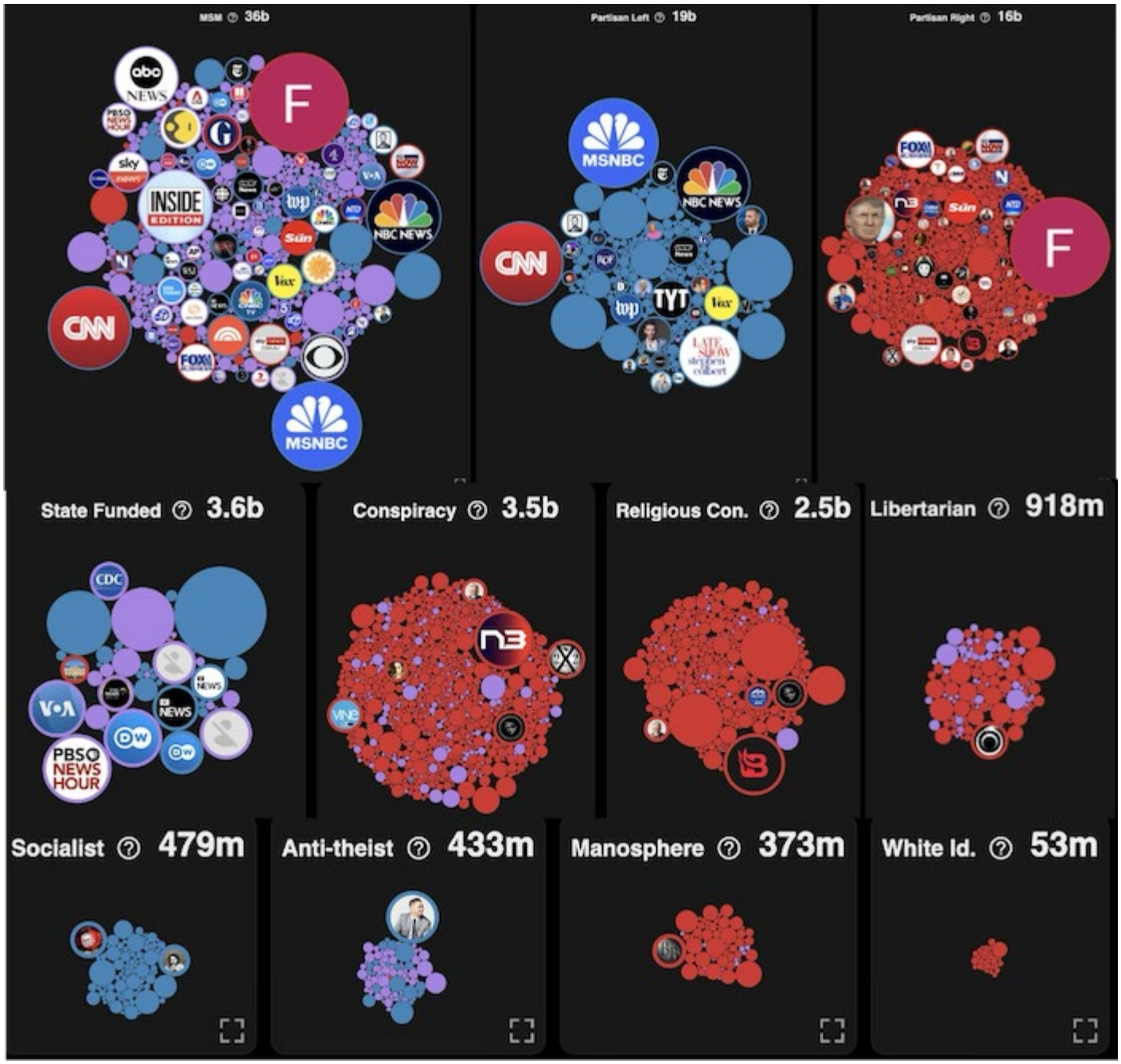

Figure 5: Visualization of soft tags.

Figure 6 shows that the Recfluence dataset already captured most of the subscriptions received by "Partisan Left" content; the new dataset does not add many subscriptions to this category. However, the smaller channels from "Partisan Right" and "Conspiracy" increase significantly when we include the new channels. The Recfluence dataset accounted for only 15 percent of total subscribers for the "Conspiracy" category. When we add the new subscriptions and compare the two main categories and when all smaller channels are accounted for, we can observe that "Partisan Right" channels have a more significant number of total subscribers than "Partisan Left."

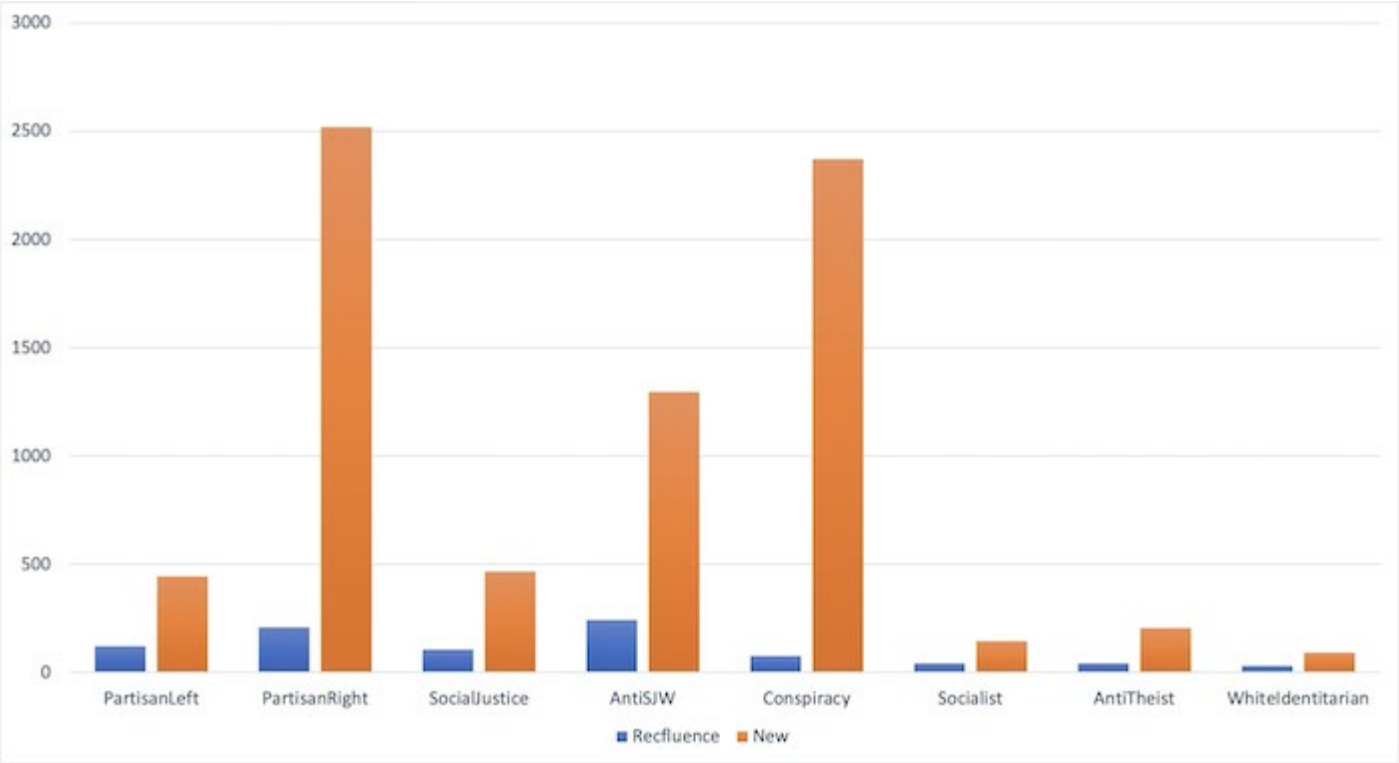

Figure 6: Number of Recfluence and new channels.

Based on our dataset, we can also identify which soft tag categories had the most significant increase in new channels found compared to the Recfluence data. Figure 7 shows that the "Conspiracy" and "Partisan Right" categories have a significant number of newly found subscriptions. The majority of these new subscriptions are for smaller channels.

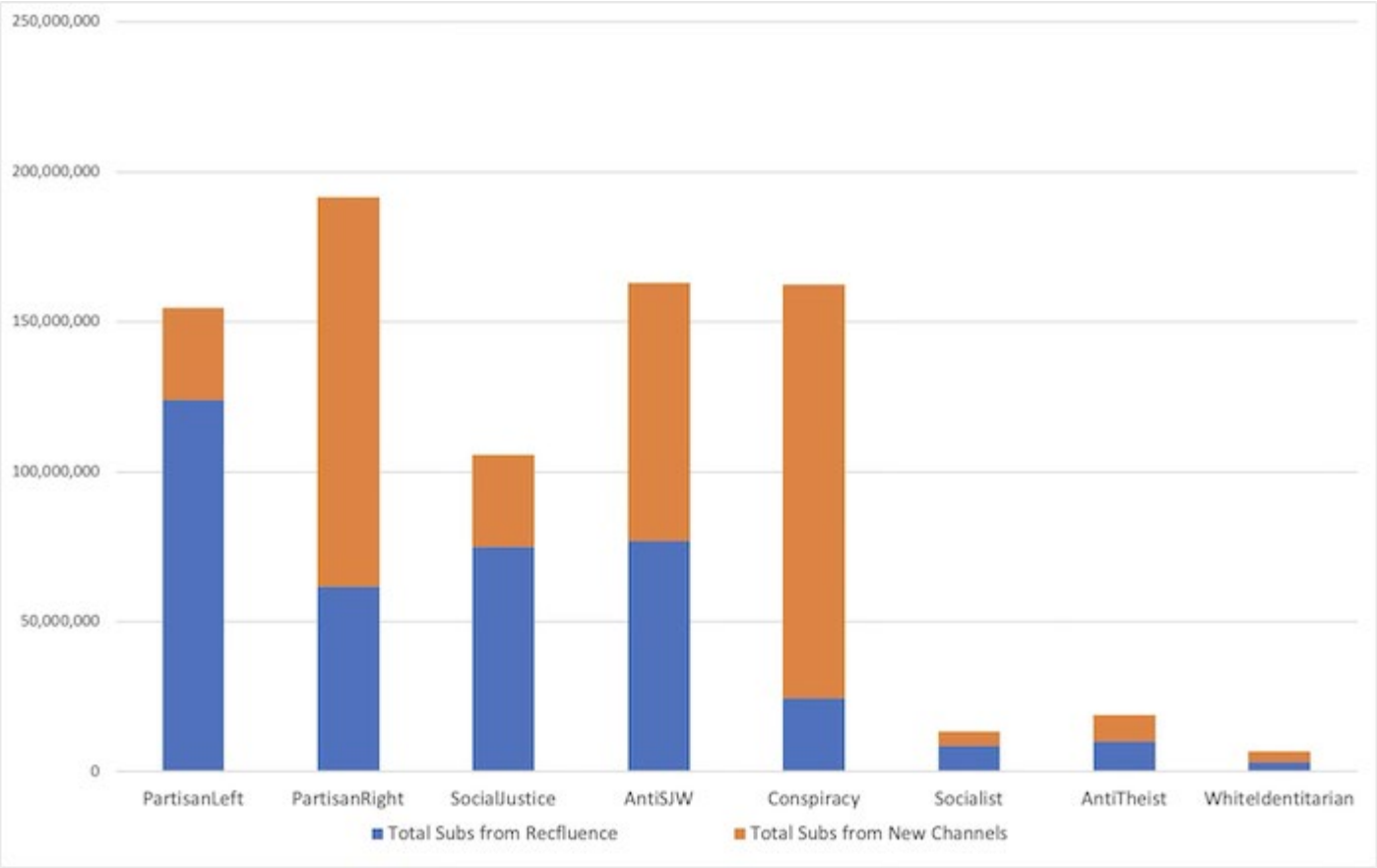

Figure 7: Subscriptions by soft tag for Recfluence and new channels.

1. Overlapping channel tags

Given that a channel can and frequently does have multiple soft tags, we decided to first look at the overlap of groups represented by soft tags. This can be seen in Figure 9, which shows the percentage of channels with a soft tag in the leftmost column that also has a soft tag in the topmost row. It is important to note that it is not symmetric due to some soft tags being much more common than others. For example, 76 percent of "White Identitarian" channels are also "Partisan Right," but only four percent of "Partisan Right" channels are "White Identitarian."

An interesting finding is a distinction between the "Partisan Right" and "Partisan Left." Where 42 percent of Partisan Left channels are also "Social Justice" channels and vice versa, the "Partisan Right" category overlaps with "AntiSJW" and "Conspiracy" channels. These smaller categories share no overlap, so they are strictly stratified between Left and Right. This phenomenon could be seen as an example of polarization not only from a traditional left-right political perspective but also from a cultural perspective as well. The overlap between "Conspiracy" and "Partisan Right" is also quite striking, with over one-third of the channels in each group belonging to the other.

	Partisan Left	Partisa n Right	Social Justice	AntiSJW	Conspiracy	Socialist	Anti Theist	White Identitarian
Partisan Left	100%	0%	42%	0%	0%	8%	1%	0%
Partisan Right	0%	100%	0%	24%	34%	0%	0%	4%
Social Justice	42%	0%	100%	0%	0%	21%	2%	0%
AntiSJW	0%	42%	0%	100%	1%	0%	1%	6%
Conspiracy	0%	38%	0%	1%	100%	0%	0%	0%
Socialist	24%	0%	62%	0%	0%	100%	0%	0%
AntiTheist	2%	0%	4%	7%	0%	0%	100%	0%
White Identitarian	0%	76%	0%	66%	0%	0%	0%	100%

Figure 8: Asymmetric soft tag category overlap.

2. How representative is the head?

One common method for analyzing YouTube is to focus on the most popular channels, also known as head channels. This was done explicitly in a Pew study (Stocking, *et al.*, 2020) and indirectly by Ledwich and Zaitsev (2020) through their method of channel discovery. We next attempt to determine what coverage can be obtained by these head channels and how representative they are of all socio-political YouTube content. We achieve this by comparing the 237 channels with over 500,000 subscribers to the 8,044 channels with between 10,000 and 500,000 subscribers in Figure 9.

This chart illustrates that focusing solely on the most popular (or 'head') channels produces a skewed view of the socio-political YouTube ecosystem. For head channels alone, it appears that "Partisan Left" has nearly twice the traffic of "Partisan Right," and "Conspiracy" channels are a very distant fifth. While looking at views from all channels shows that "Partisan Left" and "Partisan Right" have quite a close level of aggregate views, and that "Conspiracy" is quite close to "Social Justice."

Prior studies have shown that right-wing content or conspiracy theorists have managed to capture a portion of the audience (Ribeiro, *et al.*, 2020; Faddoul, *et al.*, 2020). Conversely, the Recfluence dataset and study showed that larger YouTube channels, especially more prominent mainstream media channels, were often classified as left-wing (Ledwich and Zaitsev, 2020). Our analysis reveals that these two categories are, in fact, closely related in terms of the number of views, even when smaller channels are also taken into account.

An interesting feature of this dataset is the low number of views given to independent Partisan Left media, the tail channels. Based on Figure 8 and more detailed numbers presented in the Appendix, we can see that there are 56 "Partisan Left" head channels and 51 "Partisan Right" head channels with over 500 thousand subscriptions, but for the tail channels, the numbers are more disparate; "Partisan Left" group adds 1,006 tail channels to the 56 head channels, whereas "Partisan Right" group has three times as many tail channels, 3,115. Where the "Partisan Left" dominates the head channel views, the smaller "Partisan Right" channel views equalize the viewership numbers.

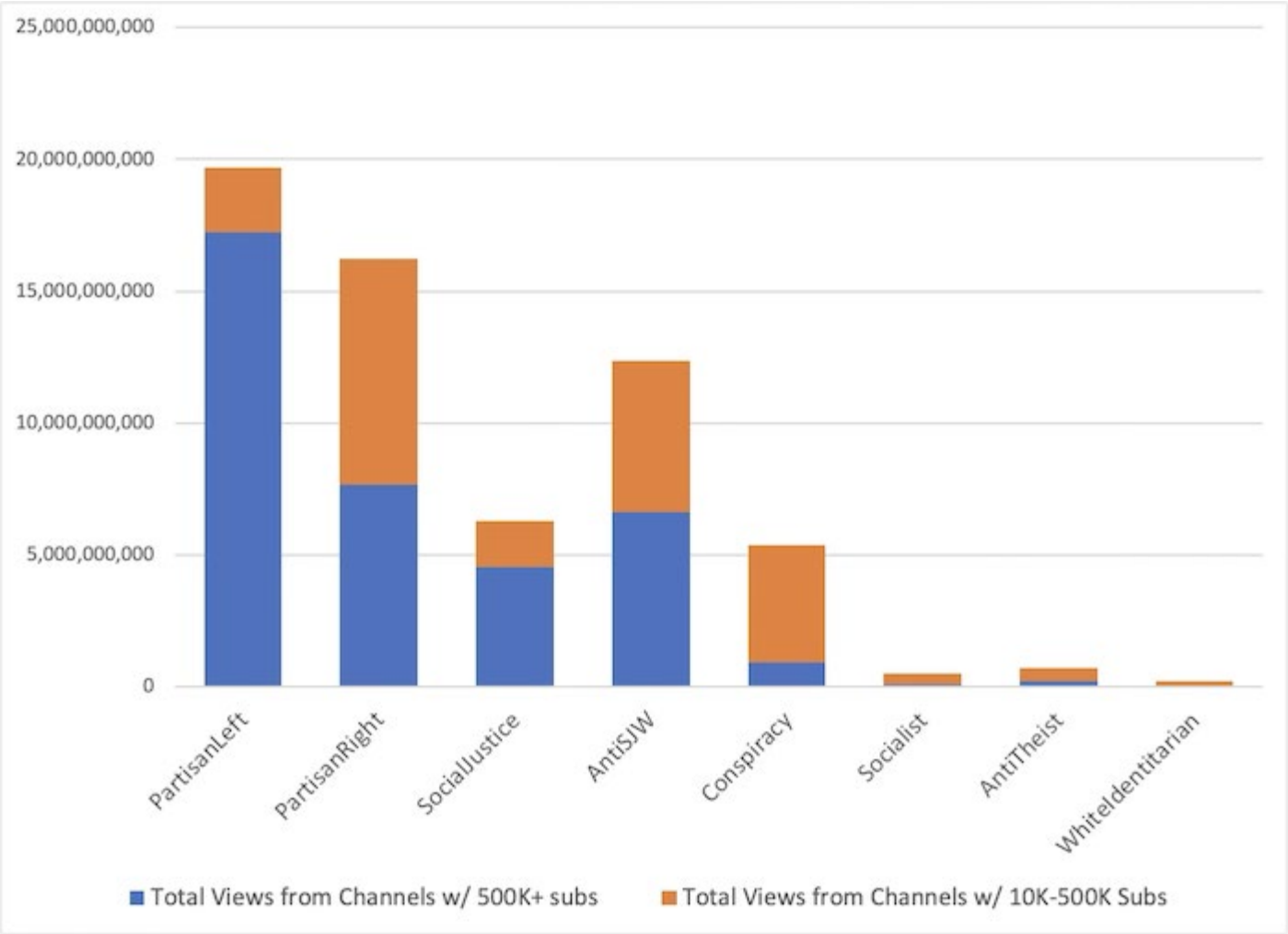

Figure 9: Head vs. tail channel aggregate views by soft tag.

However, the pure viewership data provides no reason for the differences between the head and tail communities. One can only speculate what could cause this disparity between the political spectrum opposites. The data show an overlap between the "Partisan Left" and mainstream media (Figure 10), indicating that mainstream media caters more to audiences interested in Partisan Left content. As we can see from the data, the other, more left-wing categories, such as "Social Justice" and "Socialist," are targeting a more niche audience than the mainstream "Partisan Left." On the contrary, it seems that the mainstream media offerings for the "Partisan Right" do not satisfy the audience's needs; however, viewers are eager to engage with smaller content creators. The data show that 2,569 channels are classified as "Partisan Right" with fewer than 500,000 subscriptions (as seen in Figure 10).

3. YouTube versus mainstream media

We also see striking differences in socio-political soft tags when breaking out traffic by media type in Figure 10. Most "Partisan Left" views come from "Mainstream Media" channels, while most "Partisan Right" views come from independent YouTube channels. Even more striking, there are no "Mainstream Media" channels that have "Conspiracy," "Socialist," "Anti-Theist," or "White Identitarian" soft tags. This speaks to, for better or worse, the larger breadth of ideas supported on YouTube than in "Mainstream

Media."

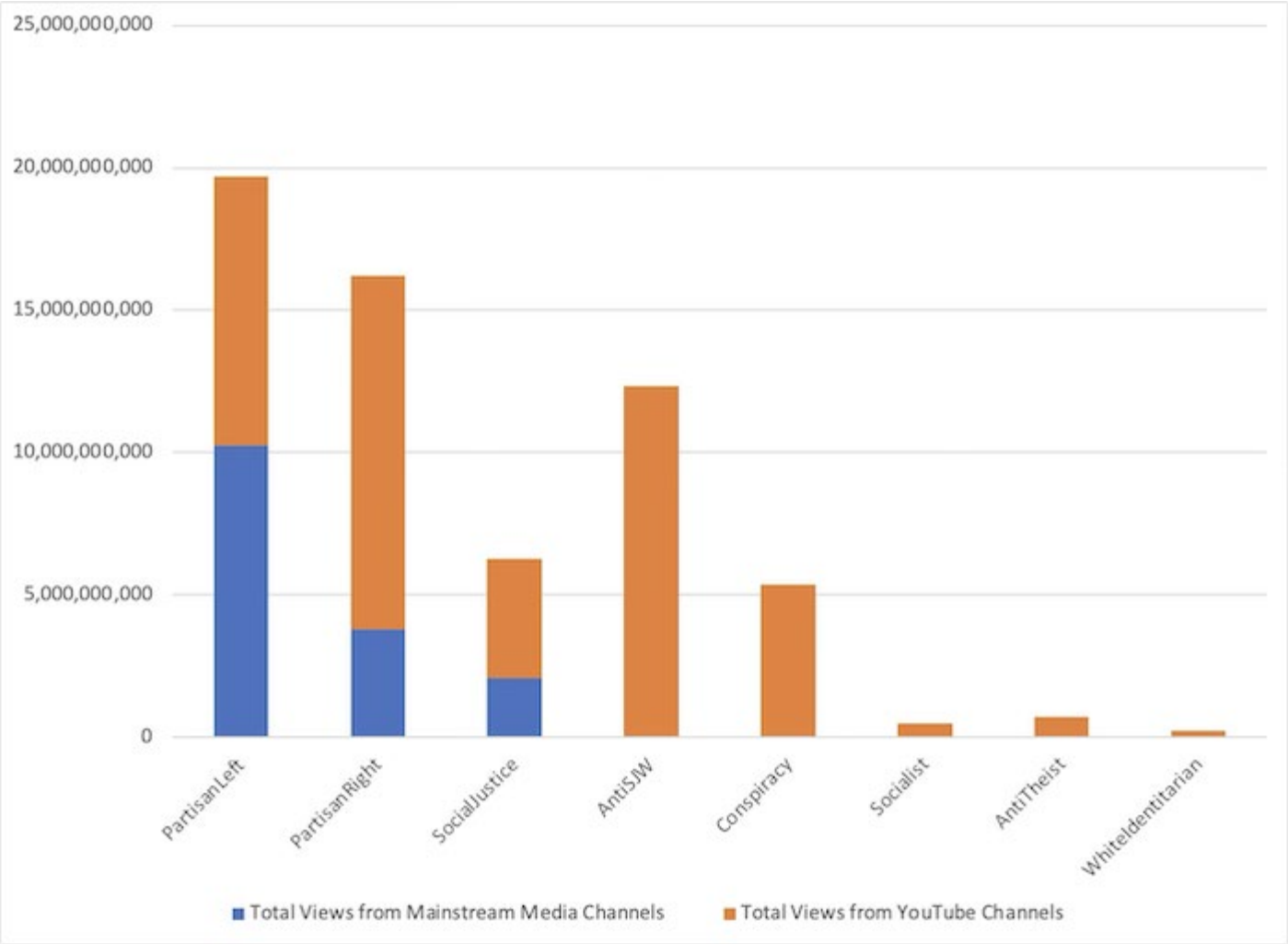

Figure 10: Mainstream media vs. independent YouTube creators' views by soft tag.

4. Yearly view trends

Another advantage of our channel discovery method is that it can identify channels that have not recently uploaded videos but still have a substantial number of subscribers. This, along with the fact that we can pull up to three years' worth of monthly channel view data from Social Blade (Social Blade, 2019), allows us to examine recent trends in views. Figure 11 shows the percentage of overall socio-political yearly views each soft tag covers.

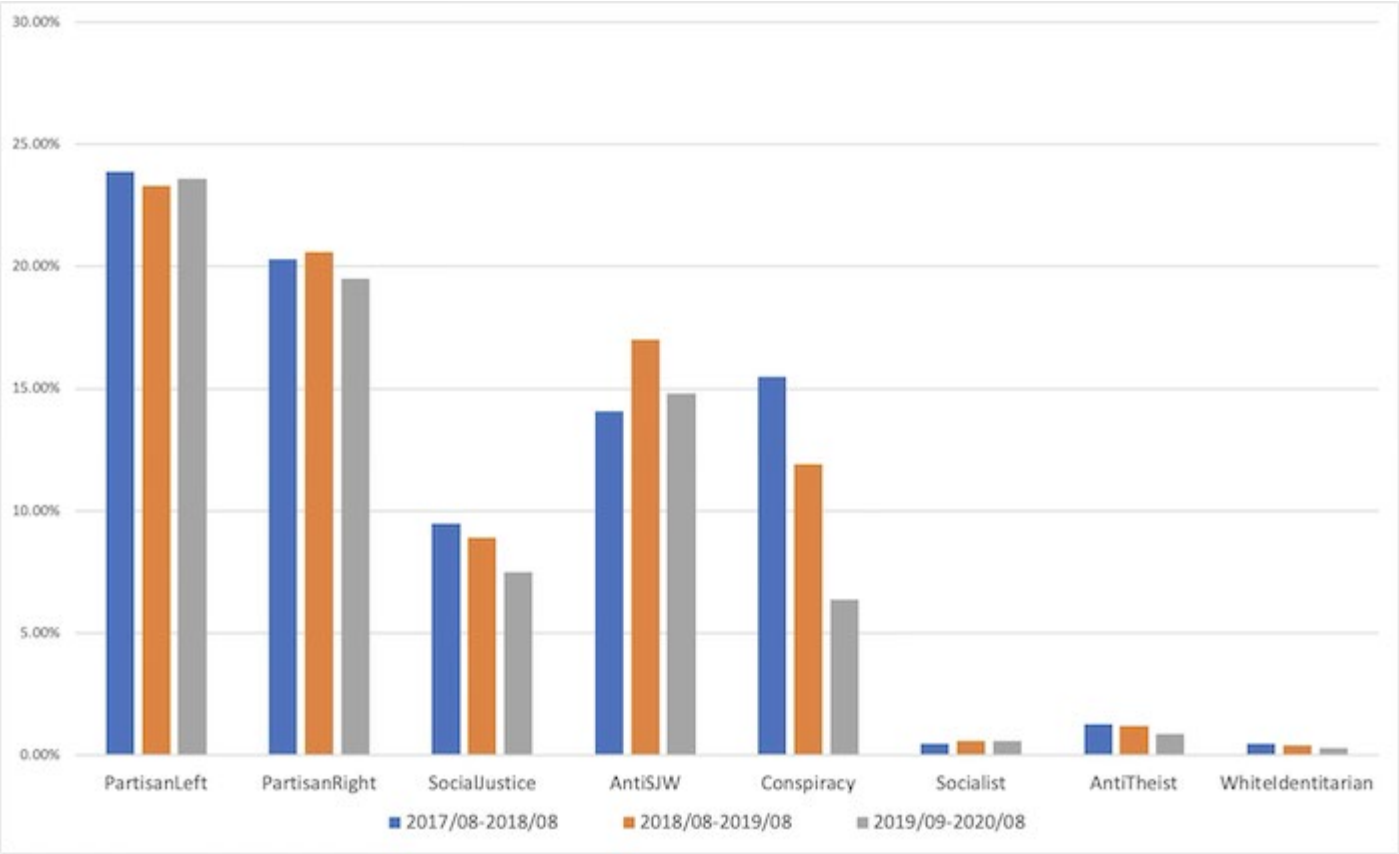

Figure 11: Yearly views per soft tag category, 2017–2020.

"Partisan Right" and "Partisan Left" have accounted for roughly the same percentage of political traffic over the last three years. We can see that most categories have experienced similar viewership numbers over the past three years. Furthermore, we can also observe that the viewership figures for political YouTube channels have plateaued. There are several notable increases in each category.

We can see that most soft tags have had a consistent percentage of overall socio-political traffic over the last three years. The one major exception is "Conspiracy," which has declined from 15.5 percent to 6.4 percent. This pattern is consistent with earlier research suggesting that YouTube's recommendation algorithm reduces exposure to conspiratorial content over time (Ledwich and Zaitsev, 2020; Faddoul, *et al.*, 2020). However, we can also observe that a significant number of viewers continue to engage with conspiratorial materials.

VI. Conclusions

This study makes several contributions to the research space on YouTube. We present new methods for automatically classifying and discovering YouTube channels, which help address the challenge of analyzing communities and categories at the platform's scale. Due to the abundance of political YouTube datasets, we focus on this category to test our methods, but we believe these methods can be applied to various other categories.

Multiple experiments are conducted to evaluate the performance of the chan2vec-knn channel

classification. Using the Recfluence dataset (Ledwich and Zaitsev, 2020), we find that chan2vec-knn can predict "soft tags" for channels more accurately than the average individual human annotator for 11/12 tags in the experiment. Additionally, seven of these tag fewer than 50 examples in the dataset, demonstrating that the method can perform quite well with a limited amount of labeled data. We can also compare chan2vec-KNN to the machine learning approach by Dinkov, *et al.* (2019) for channel classification. Chan2vec-knn shows a sizeable improvement over their model, achieving an accuracy of 83.8 percent (compared to 73.0 percent) on predicting the political lean of news channels. Since chan2vec-knn relies on commenter subscriptions, it cannot make predictions for 79 of the 412 channels in the Dinkov, *et al.* dataset; however, it can make predictions for 265 of the 270 channels with at least 1,000 subscribers. Furthermore, the channels supported by chan2vec-knn cover 99.98 percent of the total subscribers for channels in the dataset.

A single experiment is conducted to test the performance of our channel discovery method. We use the technique to identify socio-political channels that match the criteria for the Recfluence dataset. The method discovers 7,224 such channels, which is nearly 10 times the 758 channels in the original dataset. Unlike all other studies we have encountered, we go a step further and attempt to measure the percentage of all socio-political channels that our new set of channels, in addition to the original dataset, covers. Using an analysis where we hold out channels from the original dataset when running the discovery method, we estimate that our new dataset covers 78 percent of channels that match the criteria. Due to the limitations of this analysis, however, we believe the coverage might be even higher.

These newly discovered channels are then used in a brief analysis to demonstrate the importance of conducting comprehensive studies of YouTube categories, rather than focusing solely on the most popular channels within a category. By applying soft tag classifications to the newly discovered channels and integrating them with the original Recfluence dataset, we create a more comprehensive picture of the socio-political content landscape on YouTube. The 257 channels with 500,000 or more subscribers, which we refer to as "head" channels, are then compared to the entire set of channels. We find that only 1.8 percent of traffic to "head" channels is directed to "Conspiracy" channels, while 6.5 percent of overall socio-political traffic is directed to "Conspiracy" channels. Similarly, 14.6 percent of "head" traffic is directed to "Partisan Right" channels, while this figure rises to 19.8 percent when considering overall traffic. This means that a study that only analyzed the "head" socio-political channels would significantly underestimate the amount of traffic going to "Conspiracy" and "Partisan Right" channels.

VII. Limitations and future work

Some limitations to our study must be considered. First, as noted in the data collection section, YouTube appears to be phasing out the visibility of subscriptions, limiting the availability of the data for future researchers. YouTube has also introduced new features, such as channel memberships, which are not accessible on people's public profiles. There are still data linked to user accounts that could be applied similarly, such as playlists, but the accuracy might suffer. The channel content creators also have access to subscriber information, albeit with some limitations, but this data is not publicly available. Since replication of this same method depends on API access to publicly visible subscriptions, a feature that has since become more restricted on YouTube, future researchers may need to explore alternative or supplementary signals for scalable discovery.

Furthermore, like many other YouTube researchers, we analyze content at the channel level instead of the video level. This approach functions for the majority of content, which comes from channels with narrow themes. However, there are some channels with a broad range of content that are not well-suited to this approach.

Third, our method relies entirely on subscribers to determine embeddings and embeddings to classify the

type of content that a channel has. Our experiments have shown that embeddings near each other tend to have similar content. Still, there may be many tasks in highly correlated types of content would be easily misclassified. Our method is a variation of snowball sampling, which is associated with certain potential biases. For example, the method may be biased due to selection bias, *i.e.*, the data may be influenced by the seed (Naderifar, *et al.*, 2017). However, this bias can be mitigated by selecting a variety of seed channels, as was done in our data collection.

Finally, our method can only be applied to channels with a minimum number of subscribers. More experimentation is needed to determine the exact threshold, but it is improbable that channels with fewer than 100 subscribers will be discovered or classified correctly.

For brevity, we limited our approach to commenter subscriptions; however, we believe there is room for improvement by leveraging additional data. Given the significant differences between the approaches taken by Dinkov, *et al.* (2019) and others and those of chan2vec, we believe that ensembling our methods can easily lead to an improvement in performance. FM

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Appendix

A. Channel discovery algorithm

We identify channels with property P, we first establish a set of channels labeled as having property P (positive) or lacking it (negative), termed set L. During the channel discovery process, we maintain a set of discovered channels deemed "candidates" for property P, referred to as set C. As an initial step, we include all channels from L into C, despite their known classifications.

We then employ an iterative method (detailed in Algorithm 1) to query commenter subscription data for unqueried channels in C. These candidates are denoted Ci, with their subscriptions as Si (where i indicates the iteration). We aggregate all prior subscription data into set S (initially S = S1, then S = S1 + S2 + ... + Si in subsequent iterations). The chan2vec process generates embeddings E for channels in S with a minimum commenter subscription threshold, yielding embeddings for far more channels than in C, reflecting the broad subscription habits of commenters.

For channels with embeddings in E, we apply KNN using the labeled set L to predict property P. Channels predicted positive are added to C. The iteration ceases if minimal new channels join C; otherwise, it continues.

Upon completion, a final prediction set is required, outlined in Algorithm 2. We utilize the embeddings E from the last iteration, applying KNN to predict property P for channels in C2 through Cn (n being the final iteration, excluding C1, which mirrors L). Only those predicted positive are included in the final discovered set D.

```
C_1 ← C
i ← 1
repeat
    S_i ← querycommentersubs(C_i) S ← S_1 ∪ S_2 ∪ ... ∪ S_i
    E ← chan2vec(S)
    C_{i+1} ← ∅
    for x ∈ E do
        if x channel ∉ C then
        y ← chan2vecknn(L, E, x)

        if y = positive then
            C_{i+1} ← C_{i+1} ∪ {x channel}
    if |C_{i+1}| > τ
    C ← C ∪ C_{i+1}
    else
        stop process
    i ← i + 1
```

Algorithm 1: Channel discovery algorithm.

$D \leftarrow \varnothing$

for $x \in E$ do

  if x channel $\in$ C **and** x channel $\notin C_1$ **then**

    y $\leftarrow$ chan2vecknn(L, E, x)

    if y = positive **then**

      D $\leftarrow$ D $\cup$ {x channel}

Algorithm 2: Final prediction.

B. Generating embeddings

To use chan2vec embeddings for classification, we must first gather commenter subscriptions for channels in a labeled dataset and the unlabeled channels we would like to classify. We do this using the method described in the previous section.

There is no guarantee that this will result in enough subscriptions collected for each channel to generate embeddings for them. However, in our experimentation, detailed later in Section D. Classifying Political Leaning from MBFC Dataset, we found that this method can be used to find 67 percent of channels with 1,000+ subscribers and 87 percent with 10,000+ subscribers. In that experiment, we also introduced alternative methods to increase coverage.

After collecting the data, we then use chan2vec to generate embeddings. We then use these embeddings, K-nearest-neighbors (KNN), and a dataset of labeled channels to make predictions. Specifically, if our labeled dataset consists of channels, each given a label of "1" if it has some descriptor D or "0" if it does not, and we want to predict whether channel C has descriptor D, we find the K closest channels in the labeled dataset to C using our chan2vec embeddings. Our prediction for C is the average label of the K closest channels. We use a K value of 5 or 10 for most of our experiments, but there is room for more experimentation here. The threshold for the number of nearest neighbors that must have a positive label before making a positive prediction also varies by experiment.

C. Channel discovery process

Next, we use a dataset of 773 non-political channels (Laukemper, 2020) as negative instances for socio-political classification, called the Laukemper dataset. Channels with over three million subscribers, not included in the Recfluence set, are also assumed to be negative (*i.e.*, not politically oriented content), given Recfluence's comprehensive discovery of popular channels. The channel discovery process runs for four rounds with: K=10 for KNN, cross-validation to optimize chan2vec-knn precision (≥0.9 recall) using latest embeddings and adding queried negative channels to the labeled dataset.

Final predictions use 200-dimensional embeddings from the last round, plus 16-dimensional embeddings, with averaged KNN results for improved accuracy and smoother precision/recall. Channels with fewer than

20 commenter subscriptions are filtered out, aligning with Recfluence's minimum. Table II presents the final model performance using a KNN threshold of 0.8.

Table II: Model performance.		
	Laukemper + Recfluence	**Laukemper + Recfluence + heuristic negative channels dataset**
Base rate	0.501	0.126
AUC	0.989	0.996
Precision	0.988	0.984
Recall	0.902	0.902

Table III shows the results from each round of this process. "New Positive Channels" is the number of channels identified as candidate channels during the given round that the final model ultimately predicted to be positive. "Tot Subs for New Positive Channels" is the total number of subscribers for all these channels. In the first round, these statistics only cover channels from the Recfluence dataset.

Table III: Channel discovery.				
Channel discovery round	**Total new candidate channels**	**New heuristic negative channels**	**New positive channels**	**Total subs for new positive channels**
1	1,449	0	758	431,550,776
2	16,153	3,643	5,259	489,726,779
3	8,965	790	1,732	115,027,796
4	3,079	25	233	33,284,599

Despite only running for four rounds (including the initial round), this process discovers a very large number of channels:

- 12,600,000 total channels
- 1,600,000 with at least the 5+ commenter subscriptions necessary for chan2vec
- 600,000 of these with ≥ 10,000 subscriptions (the threshold we set for the experiment)
- 25,000 of these were identified by the discovery process as candidate socio-political channels
- 7,000 of these are predicted to be socio-political by the final

This immense number of channels shows why a machine-learning approach like chan2vec-knn is necessary for classifying channels discovered through commenter subscriptions.

To evaluate model performance, we use standard classification metrics such as precision, recall, and AUC, in line with prior studies. While these metrics have limitations, especially when aggregating across heterogeneous classes, we have included them to facilitate comparison with previous work and to provide

interpretable benchmarks.

We examine the precision and recall of each tag classifier, as well as metrics related to tag agreement. This includes "Reviewer Agreement," which is the percentage of cases in which all pairs of manual reviewers agreed on whether a channel should have a given tag, as well as "Model Agreement," which is the percentage of cases in which predictions by the model agreed with each manual reviewer for each channel. This metric is included to compare the performance of automatic and manual tagging; the Recfluence dataset was validated by calculating a reviewer agreement percentage (Ledwich and Zaitsev, 2020). We include the reviewer agreement as an informal benchmark, though we acknowledge this is not a standard method in content analysis.

Since YouTube and Mainstream Media tags were annotated using a source other than manual reviewers, they do not have agreement metrics. We find that the model has a higher agreement percentage for the 13 tags with agreement metrics than the average individual reviewer for 11 of them. This is partly due to the model's impressive performance and the difficulty of manual annotation for many tags, resulting in low reviewer agreement. The model's poor performance on the "Missing Link Media" and "State Funded" tags is also understandable. These tags are meant to capture the production value and funding source of a channel. Neither of these factors is likely to identify channels with similar subscribers. For example, *BBC News* and *Russia Today* are both "State Funded" but likely have different sets of subscribers.

D. Classifying channel language

YouTube boasts over two billion monthly active users, with a significant portion of content in non-English languages. Our socio-political dataset and discovery sets (Laukemper and Recfluence) focus on English channels, yet many heuristically added negative channels are non-English, leading chan2vec-knn to classify them as negative. However, numerous non-English channels remain politically active and cluster closely in the embedding space, suggesting socio-political relevance.

Before further analysis, we filter out non-English channels, a step likely useful for many use cases and an opportunity to demonstrate chan2vec-knn's versatility. We leverage YouTube API data, which provides default language for about five percent of channels, collecting this for 17K channels and labeling "en" and "en-GB" as English, with others as non-English. Using round four embeddings and this new labeled dataset, we apply chan2vec-knn.

With hold-one-out cross-validation and a KNN threshold of 0.5, the model yields:

- Base Rate = 0.630
- AUC = 0.937
- Precision = 0.859
- Recall = 0.946

Testing the Recfluence set, limited to English-positive channels, only two of 758 were predicted non-English (both at the 0.5 threshold), featuring foreign communist anthems, which we deem acceptable classifications. As Recfluence offers only positive examples, it measures recall but not precision, showing higher recall than the training set, possibly due to mislabeling in the default language data, a hypothesis reserved for future study

Applying the model to 7,587 socio-political channels, we identify 362 as non-English and remove them (~5 percent), a worthwhile effort given their large subscriber bases, which could skew further socio-political analysis.

E. Channel discovery performance analysis

The method used for discovering channels has two distinct parts in which predictions are made. The first is a high-recall model with the goal of casting a wide net and finding as many candidate channels as possible

but avoiding collecting too much unnecessary data. The second part is a high-precision model where we use all the commenter subscription data we have collected to make as accurate a prediction as possible about which candidate channels are positive (in this case, socio-political).

We measure precision and recall to understand how well each method is performing. Recall is calculated for each method, but precision is measured only in the second part.

a. Out-of-sample channel analysis

To better understand the performance of part two, our final socio-political classification model, we have three reviewers manually label a random sample of 100 of the 19,421 English candidate socio-political channels. After independent channel review and annotation, we reviewed the channel labels and discussed points of disagreement. During this review, we found two channels no longer had videos available. For the remaining 98 channels, we can calculate the classification metrics.

Our final socio-political classification model has the following:

- Base Rate = 0.34
- Precision = 0.82
- Recall = 0.85
- ROC-AUC = 0.924

These metrics are lower than those found on the Recfluence and Laukemper dataset (Ledwich and Zaitsev, 2020; Laukemper, 2020), but that is expected given that a much smaller proportion of channels in that dataset were challenging borderline cases due to the way that it was generated.

We also add the 100 new channels to our dataset and see how our model performs on them using hold-one-out cross-validation:

- Precision = 0.85
- Recall = 0.85
- ROC-AUC = 0.958

It is promising to see the increase in performance adding just 100 more labeled channels can produce.

b. Recfluence channel hold out analysis

The prior analysis enables us to assess the precision and recall of the final socio-political classification model for candidate channels but does not evaluate the recall of the initial step, the identification of candidate socio-political channels. Regrettably, no uniform sample of all YouTube channels exists for labeling to determine this recall. Even with such a dataset, the low proportion of socio-political channels would render the resources needed for accurate labeling insufficient.

To address this, we tested the channel discovery process by holding out Recfluence set channels to check their discovery. We applied 5-fold cross-validation, setting aside 1/5 of Recfluence channels, running the process with the remaining 4/5, and verifying if the held-out 1/5 were found.

Due to high computational demands and code complexity, we restricted this to the first round, which remains insightful as 70 percent of discoveries occur there. In this round, 92 percent of held-out Recfluence channels emerged as candidate socio-political channels. Considering additional discoveries in later rounds, we estimate recall would increase if all four rounds were tested.

Table IV: Soft tag recall comparison.

Tag	Number of channels with tags	Number of chans — candidate found	Percentage chans — candidate found
YouTube	645	604	94%
Mainstream Media	79	65	82%
Missing Link	35	27	75%
AntiSJW	243	237	98%
Partisan Right	209	205	98%
Partisan Left	122	110	90%
Social Justice	106	94	89%
Conspiracy	78	74	95%
Religious Conservative	47	43	92%
Anti-Theist	44	44	100%
Socialist	44	42	96%
Libertarian	37	36	97%
Educational	33	27	82%
White Identitarian	32	32	100%
State Funded	31	27	87%

c. Full channel discovery precision and recall

We determine the final precision and recall for the socio-political channel discovery by integrating the accuracy of both process stages. Precision, which relies solely on the second stage's final positive predictions (the first stage filters out channels), matches the final model's precision of 0.85 with new training data. Recall that the product of each step's recall is 0.92 * 0.85 = 0.78. Resource constraints limit further precision and recall assessments, yet these metrics suggest the model is robust for YouTube socio-political community studies. Table III details the results, with experimental tags further explained in Table IV.

Table V lists tag multipliers and statistics, assuming that political classification and soft tag predictions are independent, although this is unlikely to be fully accurate. Without this assumption, labeling sufficient data to measure true precision and recall for soft tag predictions would be prohibitively costly. Soft tags vary in false positive and negative rates; we adjust group size estimates using model precision and recall (Table VI) via:

- Recall = Political Channel Classification Recall * Soft Tag Classification Recall
- Precision = Political Channel Classification Precision * Soft Tag Classification Precision
- Multiplier = Precision/Recall

The multiplier scales aggregate new channels, subscribers, and views to reflect false positive and negative rate differences. The multiplier was introduced to estimate annotation variance effects.

d. Classifying political leaning from MBFC dataset

In our final experiment, we compare chan2vec-knn with Dinkov, *et al.'s* (2019) approach, which uses video metadata, transcripts, and audio, on a 421-channel dataset (left, center, right) linked to Media Bias/Fact Check (MBFC) annotations. After removing nine deactivated channels, 412 remain. As chan2vec-knn requires minimum commenter subscriptions, we replicate Dinkov, *et al.'s* predictions, comparing performance only on channels chan2vec-knn can classify. We test four data collection methods (detailed in prior section) and report results in Table V, including:

- Standard method for all 412 channels.
- Method (1) plus full commenter subscriptions.
- Method (2) plus Recfluence dataset collection.
- Method (3) plus data from 10,561 socio-political channels.

Metrics include: "Number of Channels Supported" (channels with sufficient subscriptions for chan2vec-knn), "Number of Channels with 1K+ and 10K+ Subs," "Chan2vec Multi-Class Accuracy" (label from ten nearest neighbors), "Chan2vec Regression Accuracy" (weighted average, -1 to 1), "Dinkov, *et al.* Accuracy," (*i.e.*, accuracy of the Dinkov, *et al.* model predictions on the set of channels for which the chan2vec model made predictions) and "Total Subs Supported."

Table VI shows chan2vec-knn outperforms Dinkov, *et al.* across all subsets. Dinkov, *et al.* covers small channels chan2vec-knn misses, but chan2vec-knn excels with ≥10,000 subscribers:

- Model (2) covers 97 percent of ≥10,000 subscriber channels, 84 percent of ≥1,000, and 99.60 percent of subscribers.
- Model (4) achieves 100 percent of ≥10,000, 98 percent of ≥1,000, and 99.98 percent of subscribers. Adding new channel data boosts MBFC coverage by 19 percent, despite imperfect socio-political matches, due to subscriber overlap.

Table V: Tag classification performance.

Tag	Number of channels with tags	Precision	Recall	Reviewer agreement	Model agreement
YouTube	645	0.95	0.96		
Mainstream Media	79	0.69	0.76		
Missing Link	35	0.33	0.06	95.4%	94.6%
AntiSJW	243	0.78	0.83	81.2%	84.0%
Partisan Right	209	0.71	0.68	85.6%	87.2%
Partisan Left	122	0.71	0.59	87.8%	89.3%
Social Justice	108	0.78	0.83	83.0%	85.0%

Conspiracy	78	0.89	0.82	95.3%	95.9%
Religious Conservative	47	0.67	0.17	92.7%	93.4%
Anti-Theist	44	0.68	0.89	97.0%	97.5%
Socialist	44	0.90	0.80	94.2%	95.1%
Libertarian	37	0.71	0.41	93.7%	94.3%
Educational	33	0.92	0.33	91.0%	93.9%
White Identitarian	32	0.69	0.69	96.8%	97.0%
State Funded	31	0.71	0.32	99.1%	96.7%

Table VI: MBFC political lean prediction accuracy comparison.

Commenter subscription set	Number of channels supported	Number of channels with 1K+ subs	Number of channels with 1-K+ subs	chan2vec multiclass accuracy	chan2vec regression accuracy	Dinkov, *et al.* accuracy	Total subs supporter
1) MBFC channel normal subs	202	183	124	83.2	81.7	72.8	45,733,303
2) (1) + all full subs	261	227	138	85.8	85.4	70.9	46,822,049
3) (2) + all sociopolitical channel subs	278	237	140	84.2	85.3	73.0	46,879,831
4) (3) + all new political channel subs	333	365	142	81.7	83.8	73.0	47,003,925
Dinkov, *et al.* — full MBFC predictions	412	270	142			73.3	47,012,187

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